Wi-Fi Standards, Network Scanning and SSID Hiding

Question 1. List three popular Wi-Fi standards and give one key difference in speed or frequency band for each.

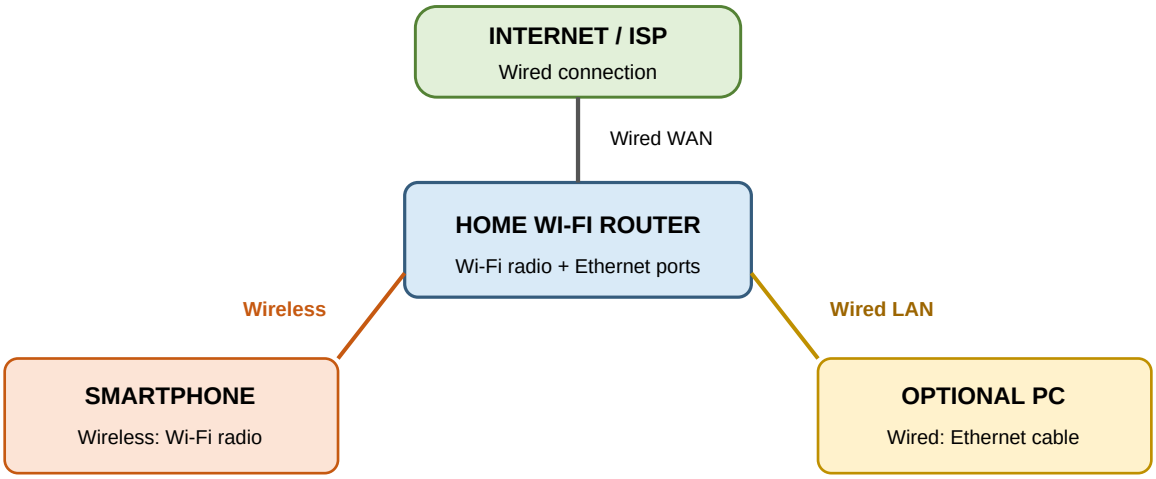
Answer: Wi-Fi standards are developed by IEEE under the 802.11 family. Newer generations generally improve throughput, efficiency, and performance in busy networks. Actual speed depends on the router, device capability, channel width, distance, interference, and number of connected devices.

Wi-Fi standard	Common generation name	Frequency band(s)	Key difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Older standard; supports up to about 600 Mbps theoretical maximum with four spatial st
802.11ac	Wi-Fi 5	5 GHz only	Designed for the 5 GHz band; theoretical rates can reach several Gbps with wide chann
802.11ax	Wi-Fi 6 / 6E	2.4 GHz and 5 GHz; 6 GHz with Wi-Fi 6E equipment	Improves efficiency in crowded networks using technologies such as OFDMA; Wi-Fi 6E

Note: These are theoretical or standard-level capabilities, not guaranteed internet speeds. Wi-Fi 6E refers to Wi-Fi 6 devices that can use the 6 GHz band; ordinary Wi-Fi 6 devices may support only 2.4 and 5 GHz.

Question 2. Draw a simple diagram showing how a smartphone connects to the internet through a home Wi-Fi router, labeling the wireless and wired parts.

Answer: The smartphone communicates with the home router wirelessly over Wi-Fi. The router then sends traffic to the internet through a wired WAN connection to a modem or ISP gateway. A computer or console can also connect to the router's LAN port with an Ethernet cable.



Solid-line connections to router are wired; smartphone-to-router link is wireless.

Step-by-step explanation

Step 1 — ISP connection: The internet service provider brings the internet connection to the home, often through fiber, cable, or another access technology. A modem or ONT may be used depending on the service.

Step 2 — Wired WAN link: The modem/ONT or ISP gateway connects to the router's WAN/Internet port using an Ethernet cable, unless the router and modem are combined into one unit.

Step 3 — Wireless LAN link: The smartphone selects the home's SSID, authenticates with the Wi-Fi password, and sends data to the router using radio signals. No physical cable runs between the smartphone and router.

Step 4 — Router forwards traffic: The router typically assigns the phone a private IP address using DHCP, then forwards internet-bound traffic toward the ISP. It commonly uses NAT for IPv4 home networks.

Step 5 — Response returns: The router sends reply traffic over Wi-Fi back to the smartphone. For a wired computer, the last link would instead be an Ethernet cable connected to a LAN port.

Question 3. Scan available Wi-Fi networks, record the SSID, signal strength and security type for at least three networks, then choose one and explain why.

Answer and practical procedure: Network names, signal strength and security settings vary by location and change over time. The table below is a **sample report only** to show the expected format; replace these example observations with the actual results from your own laptop or phone. Do not report the sample values as if you measured them.

Step 1 — Scan on a Windows laptop

1. Click the Wi-Fi icon in the Windows taskbar.
2. Wait for the list of available networks to appear.
3. Record each network name (SSID) and the displayed signal bars.
4. To inspect saved/current connection details, open Settings > Network & Internet > Wi-Fi.
5. Windows may not show security type for every nearby network in the quick list.
   For a network you are connected to, check its Wi-Fi properties or run:
   netsh wlan show interfaces
   netsh wlan show networks mode=bssid
   (Output varies by Windows version and adapter.)

Step 2 — Scan on a phone

Open Settings → Wi-Fi (or Network & Internet → Internet on many Android phones). View the nearby SSID list and signal indicator. Some phones do not display WPA2/WPA3 security for every network in the scan list; if the security type is not shown, record “not displayed” rather than guessing.

Sample observation table — replace with your actual scan results

SSID (example)	Signal strength (example)	Security (example)	Assessment
HomeNet-5G	Strong (4/4 bars)	WPA2-Personal / WPA3-Personal	Good candidate if this is your trusted home network and the password is strong.
Cafe_Guest	Medium (2/4 bars)	Open / no password	Avoid for sensitive activity unless necessary; open Wi-Fi has no Wi-Fi link encryption.
NeighborWiFi	Weak (1/4 bars)	WPA2-Personal	Encrypted, but weak signal may cause slow speed and disconnects; only connect with

Question 3 — Choosing a network and interpreting results

Step 3 — Compare signal and security

A strong signal generally improves reliability and available data rate, but signal bars alone do not prove that a network is safe. Prefer a trusted network using WPA3-Personal, or WPA2-Personal with AES, and a strong unique password. Avoid unknown open networks for banking or other sensitive work. Never attempt to access a network without the owner's permission.

Example conclusion

> “From the sample observations, I would choose HomeNet-5G because it has the strongest signal and uses WPA2/WPA3 personal security. A strong signal should give a more stable connection, while encryption and a trusted owner help protect wireless traffic. I would verify the actual SSID and security type during my own scan before submitting the report.”

What to write in your own practical record

- Date/time and location of scan (for example, home or classroom).
- Device used (Windows laptop, Android phone, or iPhone).
- At least three SSIDs exactly as displayed.
- Signal strength in bars or percentage/dBm if available.
- Security type only if shown or verified; otherwise write “not displayed.”
- Which network you chose and why, considering both signal and security.

Question 4. Explain what SSID hiding is and discuss one advantage and one disadvantage.

Answer: SSID hiding is a router setting that stops the wireless network name from being included in ordinary Wi-Fi beacon advertisements in the usual way. The network may not appear by name in a normal scan list, so a user may need to enter the SSID manually to connect. Exact behavior depends on the router and client device.

How to connect to a hidden SSID

1. Open the device's Wi-Fi settings.
2. Choose an option such as “Add network,” “Other,” or “Join other network.”
3. Type the exact SSID (network name).
4. Select the correct security type, such as WPA2-Personal or WPA3-Personal.
5. Enter the password and connect.

One advantage

Less visible in ordinary Wi-Fi lists: Hiding the SSID can reduce casual visibility because the network name is not shown like a normal advertised network in many scan lists. This may reduce accidental connection attempts by nearby users.

One disadvantage

It is not real security and makes joining less convenient: People who are allowed to connect must know and enter the exact name, and some devices may handle hidden networks less smoothly. A determined observer can still discover a hidden network by monitoring wireless traffic; hiding the SSID does not replace WPA2/WPA3 encryption or a strong password.

Best practice

Use WPA3-Personal where all devices support it, or WPA2-Personal with AES when compatibility is needed. Choose a strong, unique Wi-Fi password, keep router firmware updated, and disable WPS if it is not needed. Treat SSID hiding as an optional convenience setting, not as the main protection for a network.